

SHRI G.S. INSTITUTE OF TECHNOLOGY & SCIENCE, INDORE
INFORMATION TECHNOLOGY DEPARTMENT

Subject Code	Subject Name	L	T	P	Th. Credit	Pr. Credit	Maximum Marks				Total Credit
							End Sem	Class Work	Sessional Work	End Sem	
IT28506	COMPUTER ORGANIZATION & ARCHITECTURE	3	0	2	3	1	70	30	40	60	4

PRE-REQUISITES: Digital Logic Design.

COURSE OBJECTIVES:

1. Conceptualize the basics of organizational and architectural issues of a digital computer and Classify and compute the performance of machines.
2. Learn about various data transfer techniques in digital computer and the I/O interfaces.
3. Estimate the performance of various classes of Memories.
4. Explain the basics of hardwired and micro-programmed control of the CPU, pipelined architectures and Hazards.

COURSE OUTCOMES: After completion of course, students will be able to:

- CO1** Describe basic components and design of a computer, including CPU, memories, and input/output units.
- CO2** Apply different memory mapping techniques to improve the performance of CPU.
- CO3** Explain the instruction cycle the I/O related concepts.
- CO4** Explain the general concepts in digital logic design, including logic elements, and their use in combinational and sequential logic circuit design.
- CO5** Apply the pipelining concepts to implement the parallel processing.
- CO6** Explain the basic concept of assembly language.

THEORY:

COURSE CONTENTS:

Unit 1: Basics of Computer, Generation of Computers, Classification of Computers, Basic operational concepts, Von Neumann Architecture, Bus structures, Hardware and Software Interfaces. General register organization, Memory and Computer registers, Stack organization, Instruction formats, Addressing modes, Data transfer and manipulation, Register Transfer Language, Register transfer Bus and Memory transfer, Micro-operations.

Unit 2: Instruction codes, computer instructions, Instruction Execution, Timing and Control, Instruction cycle, Memory-Reference Instructions, and I/O and Interrupt cycle, Machine Language, Assembly Language, Assembler, Program loops, Programming Arithmetic and logic operations, subroutines

Unit 3: Input output interface, Asynchronous data transfer, Modes of transfer, Priority interrupt, Input-output and interrupt Direct Memory access(DMA), Control unit, Design of Control unit, Control Memory, Address sequencing, Binary Arithmetic, Add, Subtract, Multiply Divide Algorithms.

Unit 4: Memory Hierarchy, Memory Classification, Modern Memory Technologies. Associative Memory, Cache Memory: Cache Memory Principles, Elements of Cache Design. Mapping Functions: Direct Mapping, Associative Mapping, Set-Associative Mapping. Virtual Memory, Address Space and Memory Space, Address Mapping Using Pages, Page Replacement Algorithm.

Unit 5: Introduction to Pipelining, Basic Performance Issue in Pipelining, The Major Hurdle of Pipelining, Pipelining Hazards: Structural Hazard, Data Hazard, Control Hazard. Concepts of Risc And Cisc, Instruction Execution Characteristics, Reduced Instruction Set Architecture, Risc Pipelining, Risc v/s Cisc Controversy. Parallel Organization: Multiprocessing, Cache Coherence, Vector Computation, Parallel Processors.

TEXT BOOKS RECOMMENDED:

1. William Stallings, "Computer Organization and Architecture", 7th Edition, Pearson Education.
2. C. V. Carl Hamacher, Zvonks Vranesic, Safea Zaky "Computer Organization", 5th Edition, McGraw Hill.

REFERENCE BOOKS:

1. Morris Mano, "Computer System Architecture", 5th Edition, Prentice Hall of India.
2. S. Andrew, S. Tanenbaum, "Structured Computer Organization", 4th Edition, PHI/Pearson.
3. John P. Hayes , "Computer Architecture and Organization", 3rd Ed. McGraw Hill.

ASSESSMENT TOOLS :

Direct assessment: End-Sem Examination, Mid-Term Test, Class Assignments, Quiz, Attendance

Indirect assessment: Course End Survey

PRACTICAL:

LAB ASSIGNMENTS:

1. Introduction to assembly language programming.
2. Print the string in assembly language programming.
3. Assembly language program to implement for getting the input from user.
4. Implement the arithmetic instruction in assembly language programming.
5. Implement the logical instruction in assembly language programming.
6. Implement the assembly program for find the even and odd number.
7. Implement an assembly program to find the largest number.
8. Implement the assembly program to find the factorial of a number.